Class: Algorithms & Data Structures Semester: Spring 2025

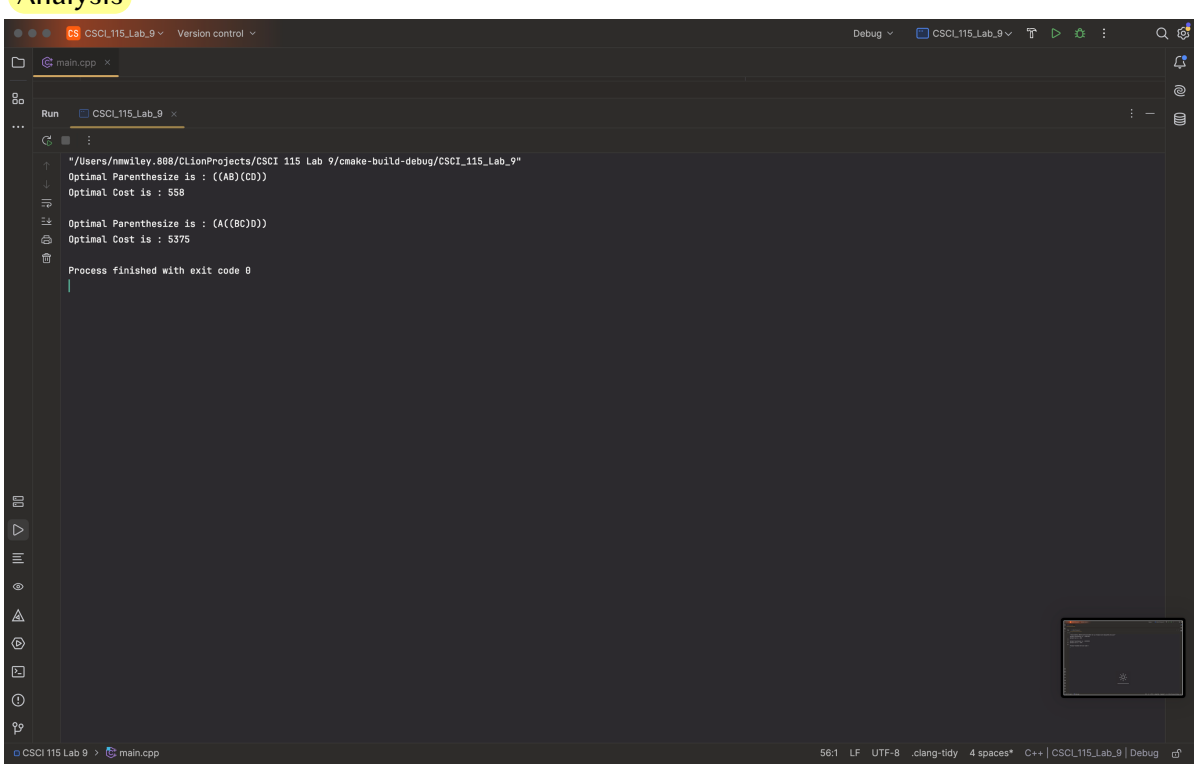
Noah Wiley
noahwiley@mail.fresnostate.edu
Lab Number - Section 09

CSCI 115 Lab 9 Report; Dynamic Programming

- State Of Objectives
  - The objective of this lab was to implement the Matrix Chain Multiplication algorithm to determine the optimal parenthesization for matrix multiplication that minimizes computational cost. The lab involved using dynamic programming techniques to compute the minimum number of scalar multiplications needed and visualize the optimal order of operations This report details the experimental procedure, analysis, encountered challenges, and conclusions drawn from this experiment.

- Experimental Procedure
  - Implemented a C++ program to solve the Matrix Chain Multiplication problem using dynamic programming
  - The program computes the minimum cost required to multiply matrices and determines the optimal parenthesization
  - Executed the program using different matrix chain dimensions to analyze performance
  - Verified the output by comparing different input cases and evaluating the results

- Analysis



- First Test Case:
  - Optimal Parenthesization: ((AB) (CD))
  - Optimal Cost: 558
- Second Test Case:
  - Optimal Parenthesization: (A(BC)D)
  - Optimal Cost: 5375
- The results indicate that the program successfully finds the optimal parenthesization for each matrix chain and calculates the corresponding multiplication cost correctly

- Encountered Problems
  - No major issues were encountered while implementing the algorithm
  - Ensured that indices were correctly handled when computing costs an splitting matrices
  - Debugging was performed to confirm the correctness of the parenthesization output

- Conclusions
  - Learned how dynamic programming optimizes matrix chain multiplication
  - Understood the significance of parenthesization in reducing computational complexity
  - Successfully implemented and tested the algorithm with different input cases

- References
  - Lab 9 - Dynamic Programming Slides